

Ramesh Sirimalla

Forward Deployed Engineer — Real-Time Data Streaming & Generative AI

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SUMMARY

Forward Deployed Engineer with 15+ years of experience building production-grade real-time data platforms, APIs, and Generative AI systems across healthcare, financial services, and public-sector programs. Experienced in architecting multi-agent workflows, enterprise RAG systems, and large-scale Kafka streaming platforms, and in working directly with business and third-party partners to take solutions into production.

CORE SKILLS

Streaming & Messaging: Confluent Kafka, Kafka Streams, Kafka Connect, KSQL, Apache Flink, Redpanda, Azure Service Bus, Spark Streaming, StreamSets

Generative & Agentic AI: LLMs, Prompt Engineering, RAG, LangChain, LangGraph, OpenAI API, Amazon Bedrock, Model Context Protocol, LLM Evaluation, Guardrails, FAISS, Pinecone, Chroma

Programming & Frameworks: Java, Python, SQL, C++, Spring Boot, Spring Integration, Spring Batch, Hibernate, PySpark, Pandas

Data, Cloud & DevOps: Oracle, SQL Server, MySQL, MongoDB, Redis, Google Spanner, Hadoop, AWS, Docker, Kubernetes (AKS, GKE), Jenkins, GitHub Actions, Splunk, Grafana

EXPERIENCE

Forward Deployed Engineer

Dec 2023 — Present

CVS Health — Dallas, TX

Real-Time Patient & Customer Notifications

- Led high-level design and development of **real-time data streaming applications** that deliver a near-real-time experience to CVS patients and customers across **SMS, email, push, and IVR**.
- Worked directly with **third-party messaging partners** to integrate their SMS, IVR, and email platforms through RESTful web services and take the integrations into production.
- Built **Kafka Streams** applications with Spring Boot that prevent duplicate messages and guarantee delivery within the **legal time window** for patient communications.

Streaming Proofs of Concept

- Built a POC decoupling RESTful services with **Azure Service Bus** for payment-transaction audits during the RxP prescription purchase cycle, with stateful computation in **Apache Flink**.
- Evaluated next-generation streaming platforms through multiple POCs on **Redpanda, Apache Kafka, and Apache Flink** across healthcare use cases.

Technical Leadership

- Lead engineering teams delivering healthcare solutions with product and business partners, and mentor junior engineers on streaming design, best practices, and technical skills.

Senior Lead Engineer

Jun 2020 — Dec 2023

CVS Health — Dallas, TX

Retail Orchestration Channel Management (ROCM) — Tier 1 Orchestration Hub

- Designed and built Spring Boot services for a **real-time compliance-check API** and a **Patient Preferences update API** serving CVS.com users.
- Built inbound RESTful services for **dispositions from third-party applications**, and integrated internal **submit-order, refill-reminder, and schedule-refill** APIs.

- Built **Spring Integration** applications on the **Confluent Kafka** API to capture closing events in real time, apply business logic, and call REST APIs of other CVS organizations to keep systems in sync.
- Built Spring Boot scheduler applications that load ROCM configuration nightly from **Oracle to MongoDB**.
- Set up **high and critical alerting with C3 and Splunk** for this Tier 1 orchestration hub.

Engineering Practices

- Led code reviews to maintain quality standards and compliance with healthcare industry regulations.

Senior Data Engineer

Jan 2019 — May 2020

Early Warning Services (Zelle) — Scottsdale, AZ

- Developed Spring Boot services with Kafka Connect and KSQL storing audit transactions for real-time dashboards.
- Designed complex KStream topologies to dedupe messages and join multiple topics into a single downstream payload.
- Built StreamSets pipelines for Mastercard reconciliation, and Grafana and Splunk dashboards for Visa and Zelle reporting.

Kafka Lead Developer

Jun 2018 — Dec 2018

CBRE — Dallas, TX

Sequentra Real Estate Data Integration

- Built the Kafka-based integration that streams data from **Sequentra**, CBRE's cloud corporate real estate platform (properties, transactions, projects, occupancy, and space management), to the external **LeaseAccelerator** lease-accounting system through Kafka messages and in-house microservices.
- Implemented complex **Confluent Kafka producer** use cases publishing **3 million records** into the Kafka cluster.
- Built **pub-sub** producer and consumer services for payments, reports, occupancies, and other use cases, updating customer applications in real time.
- Implemented **replay and retry** logic for business needs, with consumers that reset offsets using the Kafka **consumer seek API**.
- Worked closely with **data scientists**, providing the data inputs required by business requirements, and took part in application design document (ADD) discussions with the strategic architect.
- Built a proof of concept provisioning AWS infrastructure with **Terraform**, and worked on **CI/CD** pipelines with DevOps tools.

Big Data Developer

Apr 2015 — May 2018

New York State (ITS) — Albany, NY

- Integrated Kafka and Spark Streaming for real-time analytics and built big data solutions on the Hadoop ecosystem.

Senior Software Engineer

Apr 2013 — Mar 2015

New York State Police (DCJS) — Albany, NY

Senior Software Engineer

Nov 2011 — Mar 2013

Pearson VUE — Iowa City, IA

- Built the **fee-processing system** for the **NBPTS National Board Certification** program (about 55,000 certified teachers and counselors), from requirements and design documents (FSD, SDD, HLD/LLD) through **go-live and stabilization**, then rolled it out to other brands.
- Developed the core **journal-entry flow** and about **10 use cases** (fund distribution, scholarships, approvals, third-party payer setup), plus journal-entry reports and stored procedures.
- Built **CyberSource** credit-card and eCheck reconciliation, and fixed Hosted Order Page timeouts with a POST URL integration (Java, Struts, EJB, Ajax, SOAP web services).

EDUCATION & CERTIFICATIONS

Bachelor of Engineering, Computer Science — MVSR Engineering College (Osmania University), Hyderabad, India, 2001 – 2005 (First Class)

Confluent Fundamentals Accreditation — Confluent